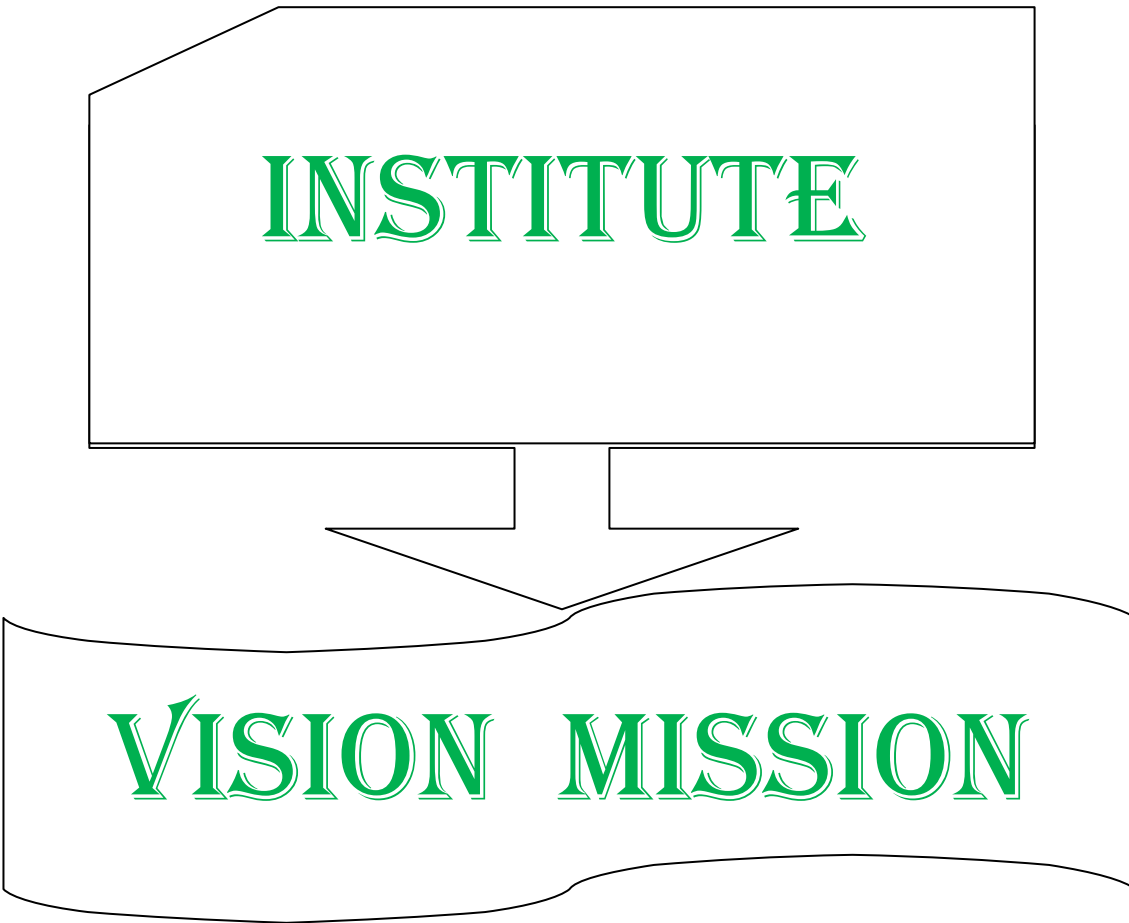


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INSTITUTE

The diagram consists of a rectangular box at the top with the word 'INSTITUTE' in green, serif, all-caps font. Below this box is a large, downward-pointing arrow. The arrow is formed by two vertical lines extending from the bottom of the box, which then curve inward and meet at a point. Below the arrow is a wide, horizontal banner with wavy, irregular edges. Inside the banner, the words 'VISION' and 'MISSION' are written in the same green, serif, all-caps font, separated by a wide space.

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. **[K3]**
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.a8@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, **TADEPALLIGUDEM** – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 10-03-2022

Revised Academic Calendar **For B.Tech VI semester , Academic Year 2021-22** **(2019 Admitted Batch)**

VI Semester			
Commencement of Next Year Class Work	07.03.2022		
I Unit of Instructions	07.03.2022	23.04.2022	7 W
I Mid Examinations	25.04.2022	30.04.2022	1 W
II Unit of Instructions	02.05.2022	18.06.2022	7 W
II Mid Examinations	20.06.2022	25.06.2022	1 W
Comprehensive Test	27.06.2022	02.07.2022	1 W
Preparation & Practicals	04.07.2022	09.07.2022	1 W
End Examinations	11.07.2022	23.07.2022	2 W
Commencement of Next Semester Class Work	25.07.2022		
Note : Calendar is prepared with 8 hrs/day hence 7 weeks per instruction period. Internal Examinations shall be conducted during the instructional days.			


PRINCIPAL

Copy to : ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: VI Semester

Section: A

Class Coordinator: Mrs. Y. Divya Vani

Room No : G-201

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	PCS-IV		CNS	CD	Lunch Break	Technical Training / TS-IV		
Tue	IOT	CD	STM	OOAD		DM	DM Lab	
Wed	CD	DM	OOAD	CNS		STM	DM	IOT
Thu	DM	Technical Training / TS-IV				OOAD	CD	CNS
Fri	STM	OOAD	OOAD Lab			IOT	CD	DM
Sat	CNS	IOT	MPS Lab			CNS	IOT	STM

Section: B

Class Coordinator: Mrs. N. Hiranmayee

Room No : G-202

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DM	CD	STM/ML	OOAD	Lunch Break	Technical Training / TS-IV		
Tue	OOAD	CNS	MPS Lab			IOT	CNS	CD
Wed	PCS-IV		STM/ML	DM		OOAD	OOAD Lab	
Thu	CD	Technical Training / TS-IV				CNS	IOT	DM
Fri	IOT	CD	DM	STM/M L		CNS	OOAD	IOT
Sat	STM/ML	DM	DM Lab			IOT	CD	CNS

Section: C

Class Coordinator: Mrs. G Prashanthi

Room No : G-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7	
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)	
Mon	OOAD	IOT	DM Lab		Lunch Break	CD	IOT	DM	
Tue	CD	Technical Training / TS-IV				PCS-IV		OOAD	
Wed	IOT	STM	CNS	CD		MPS Lab		CNS	
Thu	CNS	OOAD	DM	IOT		STM	CD	DM	
Fri	CNS	DM	IOT	CD		Technical Training / TS-IV			
Sat	STM	CNS	STM	DM		OOAD	OOAD Lab		

Section: D

Class Coordinator: Mrs. A NagaJyothi

Room No: G-204

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	CNS	STM	IOT	CD	Lunch Break	PCS-IV		OOAD
Tue	DM	Technical Training / TS-IV				IOT	CNS	DM
Wed	OOAD	DM	DM Lab			CNS	STM	CD
Thu	STM	OOAD	CD	IOT		DM	OOAD Lab	
Fri	IOT	STM	MPS Lab			Technical Training / TS-IV		
Sat	CD	DM	CNS	CD		OOAD	CNS	IOT

Staff Details:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
1.	V18CST19	Compiler Design (CD)	Mr. David Raju. K	Mr. David Raju. K	Mrs. P. Suneetha	Mrs. P. Suneetha
2.	V18CST20	Data Mining (DM)	Mrs.Y.Divya Vani	Mrs.Y.Divya Vani	Mrs. A NagaJyothi	Mrs. A NagaJyothi
3.	V18CST21	Object Oriented Analysis and Design through UML (OOAD)	Dr. D. Jaya Kumari	Mrs. G Prashanthi	Mrs. G Prashanthi	Ms.G Nagavallika
4.	V18CST22	Cryptography & Network Security (CNS)	Mrs. N. Hiranmayee	Mrs. N. Hiranmayee	Mr.,G.Sriram Ganesh	Mr.M V V Krishna
5.	V18CST23	Professional Elective-II Software Testing Methodologies (STM)	Mr.N.V.Murali Krishna Raja	Mr.N.V.Murali Krishna Raja	Mr. R L Phani Kumar	Mr. R L Phani Kumar
6.	V18ECTO E1	Open Elective – I Internet of Things (IOT)	Mr.S Harsha Vardhan	Mr.S Harsha Vardhan	Mr.S Harsha Vardhan	Mr. K N H Srinivas
7.	V18CSL08	Object Oriented Analysis and Design through UML Lab (OOAD LAB)	Dr. D. Jaya Kumari / Mrs. N. Hiranmayee / Mrs. G Prashanthi	Mrs. G Prashanthi / Mrs. N.Hiranmayee	Mrs. G Prashanthi / Mrs. D S L manikanteswari	Ms.G Nagavallika / Mrs. D Anjani Suputhri Devi
8.	V18CSL09	Data Mining Lab (DM LAB)	Mrs.Y.Divya Vani / Mrs. P. Suneetha	Mrs.Y.Divya Vani/ Mr. B Kiran Kumar	Mrs. A NagaJyothi / Dr. P Laxmikanth	Mrs. A NagaJyothi / Mr. B Kiran Kumar
9.	V18CSMPS	Mini Project with Seminar(MPS)	Mr. B Kiran Kumar	Mrs. D S L Manikanteswari	Mr. M Babu Rao	Mrs. D Anjani Suputhri Devi
10.	V18ENI06	Professional Communication Skills –IV(Verbal) PCS-IV	Dr. T Sujani	Dr. T Sujani	Mr. T V S Suresh	Mr. T V S Suresh
11.	V18CST63	Technical Skills-IV	Ms. M Radha Mangamani / Mr. P Rajesh			
		Technical Training	Dr. V. Venkateswara Rao			

Lab Venues:

S.No.	Name of the Lab Course	Lab Venue
1	Object Oriented Analysis and Design through UML Lab (OOAD LAB)	James Gosling Lab
2	Data Mining Lab (DM LAB)	James Gosling Lab
3	Technical Skills-IV (A&B)	E F CODD Lab
4	Technical Skills-IV (C&D)	James Gosling Lab
5	Technical Training	Yellow Lab
6	Mini Project with Seminar (MPS)	E F CODD Lab


 Head of the Department

COURSE STRUCTURE

VI Semester

VI - Semester							
S.No.	Course Code		Course	L	T	P	C
1	V18CST19	PCC	Compiler Design	3	0	0	3
2	V18CST20	PCC	Data Mining	3	0	0	3
3	V18CST21	PCC	Object Oriented Analysis and Design through UML	3	0	0	3
4	V18CST22	PCC	Cryptography & Network Security	3	0	0	3
5	Elective - II			3	0	0	3
	V18CST23	PEC	1. Software Testing Methodologies				
	V18CST24		2. Principles of Programming Languages				
	V18CST25		3. Machine Learning				
	V18CST26		4. Image Processing				
6	Open Elective – I (Interdisciplinary)	OEC	OPE I(1-3)	3	0	0	3
7	V18CSL08	PCC	Object Oriented Analysis and Design through UML Lab	0	0	3	1.5
8	V18CSL09	PCC	Data Mining Lab	0	0	3	1.5
9	V18CSMPS	Mini Project	Mini Project	0	0	4	2
10	V18ENT05		Professional Communication Skills - III	4	0	0	MN C
11	V18CST63		Technical Skills-IV	4	0	0	MN C
Total				26	0	10	23

Total Contact Hours: 36

**LESSON
PLANS**

Compiler Design

LESSON PLAN

Academic Year: 2021-22

Year/ Semester: VI

Name of the Course: Compiler Design

Programme: B.Tech

Section: A,B,C& D

Course Code:V18CST19/ C312

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome
1	312.1	Describe the compilation process and lexical analyzer.[K2]
2	312.2	Construct top down parsing Techniques [K3]
3	312.3	Construct bottom up parsing techniques [K3]
4	312.4	Construct syntax directed translation [K3]
5	312.5	Produce intermediate code generation process and run time environments [K3]
6	312.6	Explain the code generation process.[K2]

Text Books:

1. Compilers, Principles Techniques and Tools- Alfred V Aho, Monica S Lam, Ravi Sethi, Jeffrey D. Ullman, 2nd ed, Pearson, 2007.

REFERENCES

1. Principles of compiler design, V. Raghavan, 2nd ed, TMH, 2011
2. Compiler Design, K. Muneeswaran, Oxford

Targeted Proficiency Level and Attainment level (For each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	55	55	55	55	55	55
	Level 2	50	50	50	50	50	50
	Level 1	45	45	45	45	45	45

Lecture Plan

UNIT – I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
		Dissemination of Vision, Mission, PEOs,POs,PSOs		1		PPT
1.	CO1	Describe Language Processors	K1	1	Lecture with discussion	PPT
2.		Explain the Structure of a Compiler	K2	2	Lecture with discussion	PPT
3.		Lexical Analysis: Explain The Role of the Lexical Analyzer	K2	2	Lecture with demonstration	PPT
4.		Define Specification of Tokens	K1	1	Lecture with demonstration	PPT
5.		Describe Recognition of Tokens	K1	1	Lecture with demonstration	PPT
6.		Construct Lexical-Analyzer Generator Lex	K2	2	Lecture with demonstration	PPT
Total Hours				10 Hours		

UNIT – II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO2	Define CFG	K1	1	Lecture with discussion	PPT
2.		Explain Lexical Versus Syntactic Analysis	K2	1	Lecture with discussion	PPT
3.		Explain Elimination of Left Recursion	K2	1	Lecture with discussion	BB
4.		Construct Left Factoring	K2	1	Lecture with discussion	BB
5.		Construct Top Down Parsing: Recursive Descent Parsing	K3	2	Lecture with discussion	BB
6.		Construct First and Follow	K3	2	Lecture with demonstration	BB
7		ConstructLL(1) Grammars	K3	3	Lecture with demonstration	BB
8.		Construct Non recursive Predictive Parsing	K3	2	Lecture with demonstration	BB
9.		Demonstrate Error Recovery in Predictive Parsing.	K3	1	Lecture with demonstration	BB
Total Hours				14Hours		

UNIT – III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO3	Explain Bottom Up Parser Classification	K2	1	Lecture with discussion	BB
2.		Explain Reductions, Handle Pruning	K2	1	Lecture with discussion	BB
		Construct Shift-Reducing	K3	2	Lecture with demonstration	BB
5.		Explain Conflicts During Shift Reduce Parsing	K2	1	Lecture with discussion	BB
6.		Define LR Parsing	K1	1	Lecture with demonstration	BB
7.		Explain Difference between LR and LL Parsers	K2	1	Lecture with demonstration	BB
9.		Construct LR Parsers,Items and the LR(0) automaton	K2	3	Lecture with demonstration	BB
10.		ExplainThe LR-Parsing Algorithm	K2	1	Lecture with demonstration	BB
11.		Construct SLR Parsing Tables	K2	2	Lecture with demonstration	BB
Total Hours				13 Hours		

UNIT – IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO4	Construct CLR (1)	K3	2	Lecture with demonstration	BB
2.		Construct LALR Parsing tables	K3	2	Lecture with demonstration	BB
3.		Complete Comparison of all Bottom Up approaches	K3	1	Lecture with demonstration	BB
4.		Define Semantic Analysis: Syntax Directed Definitions	K1	1	Lecture with demonstration	BB
5.		Explain Evaluation Orders for SDD's	K2	2	Lecture with demonstration	BB
6.		Explain Applications of SDT	K2	1	Lecture with demonstration	BB
Total Hours				09 Hours		

UNIT – V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO5	Explain Variants of Syntax Trees	K2	1	Lecture with discussion	BB
2.		Explain Three-Address Code	K2	1	Lecture with discussion	BB
3.		Demonstrate Control Flow	K3	1	Lecture with discussion	BB
4.		Explain Back-patching	K2	1	Lecture with discussion	BB
5.		Explain Run-Time Environments: Storage Organization	K2	1	Lecture with demonstration	BB
6.		Illustrate Stack Allocation of Space	K3	1	Lecture with discussion	BB
7.		IllustrateHeap Management	K3	1	Lecture with discussion	BB
Total Hours				07 Hours		

UNIT – VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO6	Explain Basic Blocks and Flow Graphs	K2	1	Lecture with discussion	PPT
2.		Describe Optimization of Basic Blocks	K2	1	Lecture with discussion	PPT
		Explain Peephole Optimization	K2	1	Lecture with discussion	PPT
3.		Explain Register Allocation and Assignment	K2	1	Lecture with discussion	PPT
4.		Illustrate Machine-Independent optimizations: The Principal Sources of Optimizations	K2	2	Lecture with demonstration	PPT
5.		Discuss Introduction to Data-Flow Analysis.	K2	1	Lecture with demonstration	PPT
Total Hours				07 Hours		

Total No. of classes: 60

Data Mining

LESSON PLAN

Academic Year: 2021-22

Year/ Semester: VI

Name of the Course: Data Mining

Programme: B.Tech

Section: A,B,C& D

Course Code:V18CST20/313

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome
1	313.1	Explain the concept of Data Mining and its functionalities.[K2]
2	313.2	Discuss various Data Preprocessing Techniques [K2]
3	313.3	Demonstrate Association Analysis Techniques. [K3]
4	313.4	Illustrate various Classification Techniques.[K3]
5	313.5	Demonstrate Alternative techniques for Classification [K3]
6	313.6	Use different Clustering techniques to cluster data.[K3]

Text Books:

1. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann Publishers

Reference Books:

1. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 1st Edition, Pearson Education Inc.

2. Data Mining and Analysis, Mohammed J Zaki, Wagner Meira JR, 1st Edition, Cambridge University Press

Targeted Proficiency and attainment Levels (for each Course Outcome):

COs		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	65	65	60	60	60	60
	Level 2	55	55	50	50	50	50
	Level 1	45	45	40	40	40	40

Lecture Plan

UNIT - I

S#	Cours e Outco me	Intended Learning Outcomes (ILO)	Knowl edge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction: Identify what motivated Data Mining.	K1	1	Lecture	BB/ICT
2.		State the importance of Data Mining. Describe kinds of Data on which Data Mining can be done.	K1	2	Lecture	BB/ICT
3.		Illustrate Data Mining Functionalities.	K2	2	Lecture+ discussion	BB/ICT
4.		Illustrate Major Issues in Data Mining	K1	1	Lecture	BB/ICT
5.		Explain Attribute Types and Basic Statistical Descriptions of Data	K2	2	Lecture + discussion	BB/ICT
6.		Illustrate Data Visualization techniques.	K2	1	Lecture + discussion	BB/ICT
7.		Describe Data Similarity and Dissimilarity measures	K2	2	Lecture + discussion	BB/ICT
		Total Hours		11Hours		

UNIT - II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO2	Data Pre-processing: Identify reasons for pre-processing the data.	K1	1	Lecture	BB/ICT
2.		Describe Descriptive Data Summarization.	K2	1	Lecture + discussion	BB/ICT
3.		Explain Data Cleaning techniques.	K2	2	Lecture + discussion	BB/ICT
4.		Discuss Data Integration and Data Transformation techniques	K2	2	Lecture + discussion	BB/ICT
5.		Explain Data Reduction techniques.	K2	2	Lecture + discussion	BB/ICT
6.		Discuss Data Discretization and Concept Hierarchy Generation.	K2	2	Lecture + discussion	BB/ICT
		Total Hours		10 Hours		

UNIT - III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	C03	Illustrate the concept of Association Analysis	K2	2	Lecture + discussion	BB/ICT
2.		Explain Frequent Itemset generation process.	K2	2	Lecture + discussion	BB/ICT
3.		Discover Association Rules from the given data.	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
4.		Present a Compact representation of frequent itemsets.	K1	1	Lecture	BB/ICT
5.		Use FP-Growth Algorithm for Association Analysis.	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
		Total Hours		9 Hours		

UNIT - IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO4	Classification : Describe the basic concepts of Classification	K1	1	Lecture	BB/ICT
2.		Illustrate the approach for solving a classification problem.	K2	1	Lecture + discussion	BB/ICT
3.		Explain the concept of a Decision Tree Induction	K2	2	Lecture + discussion	BB/ICT
4.		Construct a decision tree.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
5.		Describe the methods for expressing an attribute test conditions.	K2	1	Lecture + discussion	BB/ICT
6.		Identify the measures for selecting the best split.	K2	1	Lecture + discussion	BB/ICT
7.		Illustrate Tree Pruning	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
		Total Hours		09 Hours		

UNIT - V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO5	Classification : Alternative techniques Explain Bayes' Theorem	K2	2	Lecture+ discussion	BB/ICT
2.		Apply Naïve Bayesian Classification Algorithm	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
3.		Explain the concept of Bayesian Belief Networks	K2	2	Lecture + discussion	BB/ICT
		Total Hours		06 Hours		

UNIT – VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO6	Explain the Concept of Cluster Analysis.	K2	1	Lecture + discussion	BB/ICT
2.		Describe different types of Clustering and Different Types of Clusters.	K2	2	Lecture + discussion	BB/ICT
3.		Apply k-means and its variants for clustering.	K3	3	Lecture + discussion and In-class Assignment	BB/ICT
4.		Describe k-means additional issues.	K1	1	Lecture	BB/ICT
5.		Describe k-means and different types of clusters.	K2	1	Lecture + discussion	BB/ICT
6.		Identify k-means strengths and weaknesses	K1	1	Lecture	BB/ICT
7.		Express k-means as an Optimization Problem.	K2	1	Lecture + discussion	BB/ICT
8.		Apply Basic Agglomerative Hierarchical Clustering.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
9.		Discuss other Specific Techniques for Agglomerative Hierarchical Clustering	K2	1	Lecture + discussion	BB/ICT
10.		Explain Traditional Density: Center-Based Approach.	K2	1	Lecture + discussion	BB/ICT
11.		Apply the DBSCAN Algorithm.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
12.		List strengths and Weaknesses of DBSCAN Algorithm	K1	1	Lecture	BB/ICT
		Total Hours		15 Hours		

Total No. of classes: 60

OOAD THROUGH UML

LESSON PLAN

Academic Year: 2021-22

Year/ Semester: VI

Name of the Course: OOAD THROUGH UML Course Code: V18CST21/C314

Programme: B.Tech

Section: A,B,C& D

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome
1	314.1	Discuss importance of modeling. K2]
2	314.2	Describe classes and relationships. [K2]
3	314.3	Develop class diagrams and object diagrams. [K3]
4	314.4	Develop Interaction, Use case and Activity Diagrams.[K3]
5	314.5	Illustrate advanced behavioral modeling. [K3]
6	314.6	Develop component and deployment diagrams [K3]

TEXT BOOK:

1. The Unified Modeling Language User Guide, Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education.

REFERENCE BOOKS:

1. UML 2 Toolkit, Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, WILEY- Dreamtech India Pvt. Ltd.
2. Fundamentals of Object Oriented Design in UML, Meilir Page-Jones, Pearson Education.
3. Modeling Software Systems Using UML2, Pascal Roques, WILEY- Dreamtech India Pvt. Ltd. Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO 1	CO 2	CO 3	CO 4	CO 5	CO 6
Targeted Proficiency Level		75	60	75	60	60	75
Targeted level of Attainment	Level 3	75	60	75	60	60	75
	Level 2	65	50	65	50	50	65
	Level 1	55	40	55	40	40	55

Lecture Plan:

UNIT-I

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs, POs, PSOs		1	Lecture	BB
2		Introduction to UML: Introduction	K1	1	Lecture	BB
3		Importance of modeling - Principles of modeling	K1	1	Lecture	ICT
4		Object oriented modeling, Introduction to UML	K2	1	Lecture	ICT
5		Conceptual model of the UML	K2	2	Lecture	ICT+BB
6		Architecture	K2	1	Lecture	ICT
7		Software Development Life Cycle	K2	1	Lecture	ICT
		Total Hours		8 Hours		

UNIT-II

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Advanced Structural Modeling : Classes	K2	1	Lecture	BB
2		Relationships	K2	1	Lecture	BB
3		Common Mechanisms and Diagrams	K2	2	Lecture	ICT
4		Advanced classes	K2	1	Lecture	ICT+BB
5		Advanced relationships	K2	1	Lecture	ICT+BB
6		Interfaces	K2	1	Lecture	ICT+BB
7		Types and Roles-Packages.	K2	1	Lecture	BB
		Total Hours		8 Hours		

UNIT-III

S. N o.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Class & Object Diagrams: Terms, concepts	K2	2	Lecture	BB+ICT
2		Modeling techniques for Class Diagrams	K3	2	Lecture	BB+ICT
3		Modeling techniques for Object Diagrams.	K3	2	Lecture	BB+ICT
		Total Hours		6 Hours		

UNIT-IV

S. N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Basic Behavioral Modeling-I: Interactions	K2	2	Lecture	BB+ICT
2		Interaction diagrams.	K3	2	Lecture with Discussion	BB+ICT
3		Basic Behavioral Modeling-II: Use cases	K2	2	Lecture	BB+ICT
4		Use case Diagrams	K3	2	Lecture with Discussion	BB+ICT
5		Activity Diagrams	K3	3	Lecture with Discussion	BB+ICT
		Total Hours		11Hours		

UNIT-V

S. N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Advanced Behavioral Modeling: Events and signals	K2	1	Lecture	BB+ICT
2		State machines	K2	1	Lecture	BB+ICT
3		Processes and Threads	K2	1	Lecture	BB+ICT
4		Time and space	K2	2	Lecture	BB+ICT
5		State chart diagrams	K3	3	Lecture	BB+ICT
		Total Hours		9 Hours		

UNIT-VI

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Architectural Modeling: Component	K2	2	Lecture	BB+ICT
2		Deployment	K2	2	Lecture	BB+ICT
3		Component Diagrams	K3	2	Lecture	BB+ICT
4		Deployment Diagrams.	K3	2	Lecture	BB+ICT
		Total Hours		8 Hours		

Total No. of classes: 60

Cryptography & Network Security

Academic Year: 2021-22

Programme: B.Tech

Year/ Semester: IV

Section: A,B , C & D

Name of the Course: Cryptography & Network Security

Course Code:V18CST22/C315

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome
1	315.1	Describe the fundamentals of networks security, security architecture, threats and vulnerabilities. [K2]
2	315.2	Discuss the mathematical support for both symmetric and asymmetric key cryptography. [K2]
3	315.3	Discuss the concept of developing encryption and decryption algorithms. [K2]
4	315.4	Illustrate various techniques of encryption and message authentication functions.[K3]
5	315.5	Apply various Key management and Distribution techniques and its importance. [K3]
6	315.6	Demonstrate the Need of Transport level and Electronic mail security algorithms. [K3]

TEXT BOOKS:

1. William Stallings, “Cryptography and Network Security, Principles and Practices”, Pearson Education, Sixth Edition.
2. William Stallings, “Network Security Essentials (Applications and Standards)”, Pearson Education Fourth Edition.
3. Cryptography and Network Security, Behrouz A Forouzan, Debdeep Mukhopadhyay, (3e) Mc Graw Hill.

REFERENCE BOOKS:

1. Charlie Kaufman, Radia Perlman and Mike Speciner, “Network Security – Private Communication in a Public World” Pearson/PHI.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

UNIT-I

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs,POs,PSOs		1	Lecture	ICT
2		Explain Computer security concepts	K2	1	Lecture	ICT
3		Classify Security Attacks	K2	1	Lecture	ICT
4		Explain about Security Services and Mechanisms	K2	1	Lecture with Discussion	ICT
5		Explain OSI Security Architecture	K2	2	Lecture	ICT
6		Explain Model for Network Security	K2	1	Lecture	ICT
7		Classify Classical Encryption Techniques	K2	1	Lecture with Discussion	ICT
8		Classify Substitution ciphers	K2	2	Lecture	ICT
9		Classify Transposition Techniques	K2	2	Lecture	ICT
		Total Hours		12Hours		

UNIT-II

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Prime and Relatively Prime Numbers	K1	1	Lecture	ICT
2		Illustrate about Fermat's and Euler's Theorems	K3	2	Lecture	ICT
3		Explain The Chinese Remainder theorem	K2	2	Lecture	ICT
4		Explain Euclidean Algorithm	K2	1	Discussion	ICT
5		Explain Modular Arithmetic	K2	2	Discussion	ICT
		Total Hours		8 Hours		

UNIT-III

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Define Block Cipher	K1	1	Lecture	ICT
2		Describe Traditional Block Cipher Structure	K2	1	Lecture	ICT
3		Explain about Stream Cipher	K2	1	Lecture	ICT
4		Explain about RC4 Stream Cipher	K2	1	Lecture	ICT
5		Explain about DES	K2	2	Discussion	ICT
6		Explain Triple DES	K2	1	Discussion	ICT
7		Explain about AES-Structure	K2	2	Lecture	ICT
8		Describe Key Expansion	K1	1	Lecture	ICT
9		Discuss IDEA	K2	1	Lecture	ICT
		Total Hours		11Hours		

UNIT-IV

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Illustrate RSA Algorithms	K3	1	Lecture	ICT
2		Illustrate Diffie Hellman Key Exchange	K3	1	Lecture	ICT
3		Discuss Elliptic Curve Cryptography	K2	1	Lecture	ICT
4		Explain about Message Authentication code - Message Authentication Functions	K2	1	Lecture	ICT
5		Explain about Requirements and security for Message Authentication code	K2	2	Lecture With Discussion	ICT
6		Explain HMAC and Hash functions	K2	2	Lecture With Discussion	ICT
7		Explain Secure Hash Algorithm(SHA)	K2	1	Lecture	ICT
8		Explain SHA-512	K2	2	Discussion	ICT
		Total Hours		11 Hours		

UNIT-V

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Explain about Digital Signatures	K2	1	Lecture	ICT
2		Explain Authentication Protocols	K2	2	Lecture	ICT
3		Explain about Kerberos	K3	2	Lecture	ICT
4		Differentiate about Key management distribution(Symmetric &Asymmetric)	K3	1	Lecture	ICT
5		Explain X.509 digital certificate	K3	2	Lecture	ICT
6		Explain about NIST Digital Signature Algorithm	K2	1	Lecture	ICT
		Total Hours		9 Hours		

UNIT-VI

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Transport Level Security Discuss Web security Requirements	K2	1	Lecture With Discussion	ICT
2		Explain about Secure Socket Layer (SSL)	K2	2	Lecture	ICT
3		Explain Transport Layer Security(TLS)	K2	1	Discussion	ICT
4		Illustrate principle of Inclusion–Exclusion	K3	1	Lecture	ICT
5		Explain Electronic Mail Security: Pretty Good Privacy (PGP)	K2	2	Lecture	ICT
6		Demonstrate about S/MIME	K3	2	Lecture	ICT
		Total No. of classes: 60		9 Hours		

Total No. of classes: 60

Software Testing Methodologies

Academic Year: 2021-22

Programme: B.Tech

Year/ Semester: VI

Section: A,B,C,D

Name of the Course: Software Testing Methodologies **Course Code:** V18CST23/C316-E1

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

S.No.	CO No.	Course Outcome
1	316.1	Describe Software testing objectives and methodology. (K2)
2	316.2	Apply various Software testing techniques. (K3)
3	316.3	Discuss Static testing techniques for software testing. (K2)
4	316.4	Differentiate software testing and debugging process. (K2)
5	316.5	Construct test cases by understanding test suite management. (K3)
6	316.6	Explain modern software testing tools to support software testing. (K2)

Text Books:

1. Software testing Principles and Practices, NareshChouhan, 9th Edition, Oxford Publisher

Reference Books:

1. Software testing techniques - Boris Beizer, 2nd Edition, Dreamtech publisher..
2. Foundations of Software testing, Aditya P Mathur, 2nd edition, Pearson.
3. Software Testing- Yogesh Singh, cambridge.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	60	60	65	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

Unit-I

S.No .	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Introduction to OBE, Dissemination of Vision, Mission of the Dept.andPEOs,POs& PSOs of the Programme.		1	Lecture	BB
2		Describe Evolution of software testing.	K1	1	Lecture with discussion	BB
3		Identify Myths and Facts, Goals of software testing.	K1	1	Lecture	BB
4		Outline the Model for software testing.	K1	1	Lecture	BB
5		Describe the life cycle of a bug and its states ,Affect of bugs on software testing economics	K1	1	Lecture	BB+ICT
6		Classify bugs based on criticality and SDLC	K2	1	Lecture	BB
7		ExplainSoftware Testing life cycle.	K2	2	Lecture with Discussion and in class Assignment	BB+ICT
		Total Hours		8 Hours		

Unit- II

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Outline Verification and Validation activities	K1	1	Lecture	BB
2		Explain the Verification of requirements.	K2	1	Lecture	BB
3		Interpret the process to verify High level and Low level designs.	K2	1	Lecture	BB
4		Explain about code verification and Validation activities.	K2	1	Lecture	BB
5		Define Black box testing and list techniques.	K1	1	Lecture	BB
6		Apply Boundary Value Analysis for creating test cases.	K3	2	Lecture with Discussion	BB+ICT

7		Demonstrate Equivalence Class testing technique	K3	1	Lecture with Discussion and in class Assignment	BB+ICT
8		Illustrate Decision table based testing technique.	K3	2	Lecture with Discussion	BB+ICT
		Total Hours		10 Hours		

Unit-III

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define White Box Testing and criteria for Logic Coverage.	K1	1	lecture	BB
2		Illustrate Basis Path testing .	K2	2	lecture	BB+ICT
3		Explain various Loops and their Testing strategies.	K2	2	lecture	BB+ICT
4		Define Static Testing and its objectives	K1	1	lecture	BB
5		Explain about software Inspections.	K2	2	lecture	BB
6		Describe the technique of Structured Walkthroughs.	K2	1	Lecture with Discussion	BB
7		Explain about Technical Reviews.	K2	1	Lecture and in class Assignment	BB
		Total Hours		10 Hours		

Unit- IV

S.N o.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Define Regression testing and State when it is done.	K1	1	Lecture	BB
2		Describe the necessity of regression testing and its objectives.	K2	1	Lecture	BB
3		Explain Regression testing Techniques	K2	2	Lecture with Discussion and in class Assignment	BB
4		Describe the steps in Debugging Process	K1	1	Lecture	BB

5		Explain various Debugging techniques	K2	2	Lecture	BB+ICT
6		Explain guidelines for debugging and role of debuggers	K2	1	Lecture	BB
		Total Hours		8 Hours		

Unit-V

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Describe the benefits of minimizing test suite and state reasons why the test suite grows.	K1	1	Lecture	BB
2		Explain types of test case prioritization.	K2	1	Lecture	BB
3		Illustrate the coverage based test case prioritization	K3	2	Lecture	BB+ICT
4		Explain risk-based and operational profile based test case prioritization	K2	2	Lecture	BB
5		Demonstrate test case prioritization using relevant slices.	K3	2	Lecture with Discussion and in class Assignment	BB+ICT
6		Assessing the effectiveness of a prioritized test suite.	K3	2	Lecture	BB
		Total Hours		10 Hours		

Unit-VI

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Define the concept of software quality and its factors	K1	1	Lecture	BB
2		Differentiate Quality control and Quality Assurance	K2	1	Lecture	BB
3		Explain various quality metrics	K2	1	Lecture	BB

4		Illustrate ISO 9126 SQA model	K2	1	Lecture	BB+ICT
5		Explain CMM model	K2	1	Lecture with Discussion and in class Assignment	BB
6		Define automated Testing and identify the need for automation	K1	1	Lecture	BB
7		Classify Testing tools	K2	1	Lecture	BB
8		Outline Guide lines for selecting a testing tool	K1	1	lecture	BB
9		List some commercial testing tools	K1	1	lecture	BB+ICT
		Total Hours		9 Hours		

Total No. of Classes: 55

Internet of Things

Academic Year: 2021-22

Programme:B.Tech.

Year/Sem: VI

Course Code: V18ECTO1/317-OE1

Name of the Course: Internet of Things

Course Outcomes (Along with Knowledge Level):After Completing the course Student will be able to:

S.No.	CO No.	Course Outcome
1	317-OE1	Describe M2M and IOT Technologies. (K2)
2	317-OE1	Identify the layers and protocols in IOT. (K2)
3	317-OE1	Describe various communication technologies used in IOT. (K2)
4	317-OE1	Demonstrate various hardware components required for IOT applications. (K2)
5	317-OE1	Identify the cloud technologies. (K2)
6	317-OE1	Explain the applications of IOT. (K2)

Text Books:

1. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.
2. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, Cisco Press 800 East 96th Street Indianapolis, Indiana 46240 USA.
3. Vijay Madiseti and Arshdeep Bahga, "Internet of Things (A Hands-on- Approach)", 1st Edition, VPT, 2014.
4. Internet of Things: Architecture and Design Principles- By Raj Kamal, McGraw-Hill Education. 1st Edition, 2017.

Reference Books:

1. From Internet of Things to Smart Cities: Enabling Technologies - edited by Hongjian Sun, Chao Wang, Bashar I. Ahmad, CRC Press -2018.
2. Bernd Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things", ISBN 978-3- 642-19156-5 e-ISBN 978-3-642-19157-2, Springer.
3. IOT (Internet of Things) Programming: A Simple and Fast Way of Learning IOT, David Etter.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		50%	50%	50%	50%	50%	50%
Targeted level of Attainment	Level 3	60%	60%	60%	60%	60%	60%
	Level 2	50%	50%	50%	50%	50%	50%
	Level 1	40%	40%	40%	40%	40%	40%

Lecture Plan:

UNIT-I

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO1	Describe the evolution of IoT from M2M	K1	1	Lecture with discussion	PPT and ICT
2.		Discuss Architectural Overview	K1	1	Lecture with discussion	PPT and ICT
3.		Illustrate the Building architecture	K2	1	Lecture with discussion	PPT and ICT
4.		Explain Main design principles and needed capabilities	K2	1	Lecture with discussion	PPT and ICT
5.		Outline the IoT architecture	K2	1	Lecture with discussion	PPT and ICT
6.		Explain M2M and IoT Technology Fundamentals: Devices	K2	2	Lecture with discussion	PPT and ICT
7.		Explain M2M and IoT Technology Fundamentals: Gateways.	K2	1	Lecture with discussion	PPT and ICT
		Total Hours		8 Hours		

UNIT-II

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO2	Describe the Functionality of Layers in IoT	K2	2	Lecture with discussion	PPT and ICT
2.		Illustrate IoT protocols: Wireless HART	K2	2	Lecture with discussion	PPT and ICT
3.		Illustrate IoT protocols: Z-Wave, 6LoWPAN	K2	3	Lecture with discussion	PPT and ICT
4.		Illustrate IoT protocols: RPL	K2	1	Lecture with discussion	PPT and ICT
5.		Illustrate IoT protocols: CoAP.	K2	2	Lecture with discussion	PPT and ICT
6.		Illustrate IoT protocols: MQTT	K2	2	Lecture with discussion	PPT and ICT
		Total Classes		12		

UNIT-III

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Explain IoT Connectivity standard:IEEE 802.15.4	K2	1	Lecture with discussion	PPT and ICT
2		Explain IoT Connectivity technology: Wi-Fi	K2	2	Lecture with discussion	PPT and ICT
3		Explain IoT Connectivity technology:Bluetooth	K2	2	Lecture with discussion	PPT and ICT
4		Explain IoT Connectivity technology:Zigbee, LPWAN	K2	3	Lecture with discussion	PPT and ICT
5		Explain IoT Connectivity technology:5G Era.	K2	2	Lecture with discussion	PPT and ICT
		Total Classes		10		

UNIT-IV

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.		Classify various Sensors and functionalities	K2	2	Lecture with discussion	PPT and ICT
2.		Classify various Actuators and functionalities	K2	2	Lecture with discussion	PPT and ICT
3.		Infer the Radio Frequency Identification concept	K2	1	Lecture with discussion	PPT and ICT
4.		Discuss the role of various Embedded Devices for IoT-RASPBERRY PI.	K2	3	Lecture with discussion	PPT and ICT
5.		Discuss the role of various Embedded Devices for IoT-Beaglebone Black	K2	3	Lecture with discussion	PPT and ICT
		Total Classes		11		

UNIT-V

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO5	Describe the Data Collection concepts	K2	3	Lecture with discussion	PPT and ICT
2.		Illustrate the Storage and Computing Using a Cloud Platform for IoT Applications/Services.	K2	4	Lecture with discussion	PPT and ICT
3.		AWS for IoT – Amazon EC2	K2	3	Lecture with discussion	PPT and ICT
		Total Classes		10		

UNIT-VI

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.		Explain the real time applications of IoT - Smart and Connected Cities	K2	3	Lecture with discussion	PPT and ICT
2.		Explain the real time applications of IoT - Public Safety	K2	2	Lecture with discussion	PPT and ICT
3.		Explain the real time applications of IoT - Agriculture	K2	2	Lecture with discussion	PPT and ICT
4.		Explain the real time applications of IoT -Health Care	K2	2	Lecture with discussion	PPT and ICT
		Total Classes		9		

Total Number of Classes:60

OOAD Through UML Lab Programme

LESSON PLAN

After completion of this course, the students will be able to:

Academic Year: 2021-22

Year/ Semester: VI

Programme: B.Tech

Section: A,B,C& D

Name of the Course: OOAD THROUGH UML Lab Course Code: V18CSL08 /C318

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome
1	318.1	Develop OOAD and UML concepts to identify Classes, Use Cases and their relationships. [K3]
2	318.2	Develop Class diagrams. [K3]
3	318.3	Develop Use case diagrams [K3]
4	318.4	Construct Interaction diagrams [K3]
5	318.5	Develop State chart, Activity diagrams [K3]
6	318.6	Develop component and deployment diagrams [K3]

TEXT BOOKS:

1. The Unified Modeling Language User Guide, Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education.

REFERENCE BOOKS:

1. UML 2 Toolkit, Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, WILEY-Dreamtech India Pvt. Ltd.

2. Fundamentals of Object Oriented Design in UML, Meilir Page-Jones, Pearson Education.

3. Modeling Software Systems Using UML2, Pascal Roques, WILEY- Dreamtech India Pvt. Ltd.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

Lecture Plan:

S.N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Develop OOAD and UML concepts to identify Classes, Use Cases and their relationships	K3	3	Lecture With Discussion	ICT
2	CO2	Construct the Class Diagram for Library management System, Railway reservation System.	K3	3	Lecture With Discussion	ICT
		Construct the Class Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
3	CO3	Construct the Usecase Diagram for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Usecase Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
4	C04	Construct the Interaction Diagrams for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Interaction Diagrams for Hospital Management system	K3	3	Lecture With Discussion	ICT
5	CO5	Construct the Activity diagrams for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Activity diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
6	CO5	Construct the State chart for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the State chart Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
7	CO6	Construct the Component Diagram for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Component Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
8	CO6	Construct the Deployment Diagram for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Deployment Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT

Total Hours: 45

Data Mining Lab

Academic Year: 2021-22

Programme:B.Tech.

Year/Sem: VI

Name of the Course: Data Mining Lab

Course Code: V18CSL09/C319

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

C319.1	Demonstrate Data Pre-processing techniques.[K3]
C319.2	Demonstrate Association Rule Mining techniques.[K3]
C319.3	Demonstrate Classification techniques. [K3]
C319.4	Demonstrate the Clustering techniques. [K3]

Reference Books:

1. Data Mining: Practical Machine Learning Tools and Techniques, Ian H. Witten, Eibe Frank, Mark A. Hall, 3rd Edition, Morgan Kaufmann Publishers
2. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann Publishers
3. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 1st Edition, Pearson Education Inc.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4
Targeted Proficiency Level		75	75	75	75
Targeted level of Attainment	Level 3	70	70	70	70
	Level 2	65	65	65	65
	Level 1	60	60	60	60

Lecture Plan:

S.N O	Course Outcom e	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Demonstrate Data Preprocessing on predefined Weka dataset labor.arff	K3	3	Lecture With Discussion	ICT
		Create a student.arff dataset and Demonstrate Data Preprocessing on it	K3	3	Lecture With Discussion	ICT
2	CO2	Demonstrate Association rule process on predefined Weka	K3	3	Lecture With	ICT

		dataset contactlenses.arff using apriori algorithm.			Discussion	
		Create an employee.arff dataset and demonstrate Association rule process on it using apriori algorithm	K3	3	Lecture With Discussion	ICT
3	CO3	Demonstrate Classification process on student.arff dataset using j48 algorithm	K3	3	Lecture With Discussion	ICT
		Demonstrate Classification process on employee.arff dataset using id3 algorithm	K3	3	Lecture With Discussion	ICT
		Demonstrate Classification process on employee.arff dataset using Naïve Bayes algorithm	K3	3	Lecture With Discussion	ICT
		Create a customer.arff dataset and demonstrate Classification process on it using j48 algorithm	K3	3	Lecture With Discussion	ICT
4	C04	Demonstrate Clustering process on predefined Weka dataset iris.arff using simple k-means algorithm	K3	3	Lecture With Discussion	ICT
		Demonstrate Clustering process on dataset student.arff using simple k- means algorithm	K3	3	Lecture With Discussion	ICT

Total Hours: 30

Technical Skills-IV

Academic Year: 2021-22

Programme: B.Tech.

Year/Sem: VI

Name of the Course: Technical Skills-IV

Course Code: V18CST63/C319

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

S.No.	CO No.	Course Outcome
1	319.1	Demonstrate java fundamentals to solve real world computational problems. [K2]
2	319.2	Illustrate object orientated concepts in solving problems with reusability feature. [K2]
3	319.3	Apply collections on java to solve complex problems in linear time. [K3]
4	319.4	Make use of divide and conquer approach and java collections to solve problems in linear and logarithmic time. [K3]
5	319.5	Experiment with backtracking algorithm approach and java collections to solve complex problems in $O(n)$ or $O(n \log n)$. [K3]
6	319.6	Develop programs to solve optimization, NP Hard problems using java collections and dynamic programming method. [K3]

TEXT BOOKS:

1. Data structures and Algorithms made easy on Java: Narasimha Karumanchi.
2. Thinking on Java - O'Reilly.
3. Java Complete Reference.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

S.NO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Demonstrate Control Statements	K2	2	Lecture With Discussion	ICT
2		Illustrate Problem solving using Control Statements.	K2	6	Lecture With Discussion	ICT

Total Hours = 08

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Demonstrate OOP Concepts.	K2	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using OOP Concepts	K2	10	Lecture with Discussion	ICT

Total Hours = 12

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Demonstrate Divide& Conquer	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Divide and Conquer in linear time.	K3	6	Lecture with Discussion	ICT

Total Hours = 08

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Demonstrate Collections.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Collection concept.	K4	10	Lecture with Discussion	ICT

Total Hours = 12

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Demonstrate Backtracking Concept and Greedy Method	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Backtracking	K3	8	Lecture with Discussion	ICT

Total Hours =10

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Demonstrate Optimization, Dynamic Programming	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Collections and Dynamic Programming.	K3	8	Lecture with Discussion	ICT

Total Hours = 10

Total No. of Classes: 60

Professional Communication Skills - IV

Academic Year: 2021-22

Programme: B.Tech.

Year/Sem: VI

Name of the Course: Professional Communication Skills - IV

Course Code: V18ENT06/C321

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

C321.1	Express writer's tone and relevant ideas using different types of writing skills and prepare resume to show case skills and accomplishments. [K2]
C321.2	Organize thoughts in the discussions and express views without reticence and face interviews with aplomb [K3]
C321.3	Infer the meaning of the picture by thinking out of the box and speak without inhibitions. [K4]
C321.4	Demonstrate problem solving skills through the concepts of Percentages, Profit and loss, Simple Interest & Compound Interest and Allegation. [K3]
C321.5	Analyze appropriate methods of logical thinking on Ratio and Proportion, Partnership, LCM and HCF, Number System, Areas & Volumes. [K4]
C321.6	Calculate the end results of Cubes, Dice and Data Analysis, Time & Work, Time & Distance, Race & Games. [K3]

References:

1. Suneetha Mishara & Murali Krishanan, C., 2011, Communication Skills for Engineer's, Pearson publications.
2. Jain, T.S. & Gupta, , 2010, Interviews and Group Discussions, Upkar's Publications
3. Clif Ford, A. Whit Comb & Leslie E. Whit Comb, 2013, Effective Interpersonal and Team Communication skills, Wiley Publications.
4. Debra Fine, 2014, The Fine Art of small Talk, Piatkus publications
5. Alex, K., 2010, Soft skills, S.Chand Publications
6. Training & Placement cell, 2020, Work book -1 on Aptitude, Sri Vasavi Engineering College.
7. M Tyra, 2013, Magical Book on Quicker maths, BSC Publications.
8. K Kundan & M Tyra, 2009, Practice Book on Quicker Maths, BSC Publications.
9. Dr. RS. Agarwal, 2017, Quantitative Aptitude, Sultan Chand Publications
10. Dr. RS. Agarwal, 2017, A modern approach to verbal & on verbal reasoning, Sultan Chand Publications.

Hyperlinks

- <https://www.indiabix.com/>
- <https://www.campusgate.co.in/>
- <https://www.questionpaper.org/>

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

UNIT-I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	To make students understand the importance of Writing skills.	K2	1	Lecture	PPT/A.V
2		Should know the Types of Writing – Expository, Descriptive, Persuasive, Creative and Narrative Skills	K2	1	Discussion	A.V
3		Students learn the difference between the Resume and CV	K2	1	Lecture	PPT/ AV
4		Resume and Basic rules for a good Fresher's resume	K2	1	Lecture	PPT/A.V
5		Students learn Steps to make an effective resume format.	K2	1	Lecture	A.V
6		Students able to prepare a Formal Covering letter	K2	1	Lecture	PPT/A.V
		Total Hours		6 Hours		

UNIT-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Students can aware of Why GD is an important part in selection process.	K3	1	Lecture	PPT/A.V
2		The methodology, guidelines and characteristics of a successful Group discussion.	K3	1	Lecture	PPT/A.V
3		Etiquettes and Types of GDs-	K3	1	Lecture	PPT/A.V
4		Sentence starters for GD	K3	1	Lecture	PPT/A.V
5		Mock group discussions.	K3	3	Discussion	PPT/A.V
6		Students will be able to know the process of transforming from Campus to corporate.	K3	1	Lecture	PPT/A.V
7		Kinds of interviews & Screening	K3	1	Lecture	PPT/A.V
8		Face-to-Face, Panel & Skype interviews	K3	1	Lecture	PPT/A.V
9		Mock interviews	K3	4	Discussion	PPT/A.V
		Total Hours		14 Hours		

UNIT-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Know to speak on a topic spontaneously per a minute.	K4	1	Lecture	PPT/A.V
2		To speak spontaneously applying some strategies.	K4	2	Discussion	PPT/A.V
3		Speak on a topic comprehensively for a healthy debate.	K4	2	Discussion	PPT/A.V

4		To speak or To interpret on a picture and speak spontaneously	K4	3	Discussion	PPT/A.V
		Total Hours		8 Hours		

UNIT-IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Students can Calculate percentages in various situations.	K3	2	Lecture	PPT/A.V
2		Students can relate the relation between cost and selling price.	K3	2	Lecture	PPT/A.V
3		Students can compare the difference between SI & CI.	K3	2	Lecture	PPT/A.V
4		Students can choose different methods to solve problems in mixture & Alligation.	K3	1	Lecture	PPT/A.V
		Total Hours		7 Hours		

UNIT-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Students can differentiate shares according to their investments.	K4	1	Lecture	PPT/A.V
2		Students can compare the Ratio and Proportion.	K4	2	Lecture	PPT/A.V
3		Students can Explain the methods of LCM and HCF.	K4	1	Lecture	PPT/A.V
4		Students can discriminate the Areas and Volumes.	K4	2	Lecture	PPT/A.V
		Total Hours		6 Hours		

UNIT-VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Students can connect the relation between time and work.	K3	2	Lecture	PPT/A.V
2		Students can Associate the relation between Time and Distance.	K1	2	Lecture	PPT/A.V
3		Students can point out the faces of cube and Dice. Average of data.	K1	1	Lecture	PPT/A.V
4		Students can Explain the solutions of Data Analysis.	K1	2	Lecture	PPT/A.V
		Total Hours		7 Hours		

Total Number of Classes: 48